03-621 Week 10

Advanced Quantative Genetics

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Mapping Genes for Complex Traits

Genetic Distance and Mapping Genes

To review: genetic distance is calculated based on frequency of recombinant gametes

$$\text{Genetic distance} = \frac{\text{Recombinants}}{\text{Recombinants} + \text{Parentals}}$$

- 1% recombinants = 1 centiMorgan (cM)
- Everything here is based off of crossovers

Problem: We can't do this for human pedigrees

- We can't set up the most informative matings
- We lack complete genotype information
- No pure-breeding strains

Must use available pedigrees

- But progeny numbers are relatively small
- Any given pedigree is too small relative to the amount of data required to detect even moderately close linkage.
- If we do anyways and try to use a χ^2 test of significance of linkage, it would not tell us the actual map distance or the reliability of the estimate.

Sequential Likelihood Ratio Test (LOD Test)

Tests whether the hypothesis that two markers **are** linked (with recombination frequency $\theta = r, r < 0.5$) and provides a better explanation (higher probability) for a set of observed pedigrees than the null hypothesis that the markers **are not** linked ($\theta_o = 0.5$).

$$LOD(\theta) = \log_{10} \text{ of the odds for linkage with } \theta$$

Basic LOD test approach:

- Given a pedigree, look at two parents, and count the number of recombinant children. E.g., in 5 children, if 1 is recombinant, then $r = 0.2$.
- $\lambda = \frac{P(\text{Pedigree}|\theta_i = \langle\text{recombinant fraction}\rangle)}{P(\text{Pedigree}|\theta_o = 0.5)}$

What is the probability of s parentals and t recombinants in a sample if the underlying recombination frequency is θ?

- p = probability of parental = $1 - \theta$
- q = probability of recombinant = θ
- s = observed number of parental types
- t = observed number of recombinant types

From the binomial distribution:

$$P(s, t|N, \theta) = \frac{N!}{s!t!} p^s q^t$$

The $N!/(s!t!)$ part accounts for all possible birth orders.

$$\text{Likelihood ratio for linkage} \lambda = \frac{(N!/s!t!)(1-\theta_i)^s(\theta_i)^t}{(N!/s!t!)0.5^s 0.5^t} = \frac{(1-\theta_i)^s \theta_i^t}{0.5^N}$$

Note that the combinatorial component cancels out, so we don't need to bother with it, even if we don't know the birth order.

- To find LOD, $\log_1 0$ the above result.

Not all matings are informative

- A test for informative vs. non-informatic meiosis: can we tell if the gamete was recombinant?
- If we can tell: the meiosis is informative
  - Yes, gamete was recombinant
  - Or no, not recombinant
- Cannot tell if the gamete was recombinant

Is Support for Linkage Statistically Significant?

- $LOD(\theta_i) \geq 3$: "There is significant evidence for linkage with θ_i"
- $LOD(\theta_i) \leq -2$: "The true recombinant frequency is significantly greater than θ_i;;- (and may be as igh as 0.5 = no linkage)
- $-2 < LOD(\theta_i) < 3$: "The evidence of linkage is not decisive." It is necessary to collect information for more pedigrees

For example, if we have $LOD(\theta_1 = 0.2) = 0.41$. Therefore we cannot decide. We need about 7x as much additional (consistent) data to get a LOD score of 3 and make a compelling case for linkage at 20 cM. However, additional data may lead to a different estimate of θ.

How to Proceed

- Collect more pedigrees
- Re-calculate r across all the pedigrees (the new pedigrees may exhibit different recombinant fractions)
- Repeat LOD analysis on all pedigrees with the re-calculated r
- **Continue increasing sample size until clear evidence for or against linkage is obtained (LOD ≥ 3 or LOD ≤ -2)**

Further pursuit of linkage to a particular marker is not a high priority unless LOD > 0.

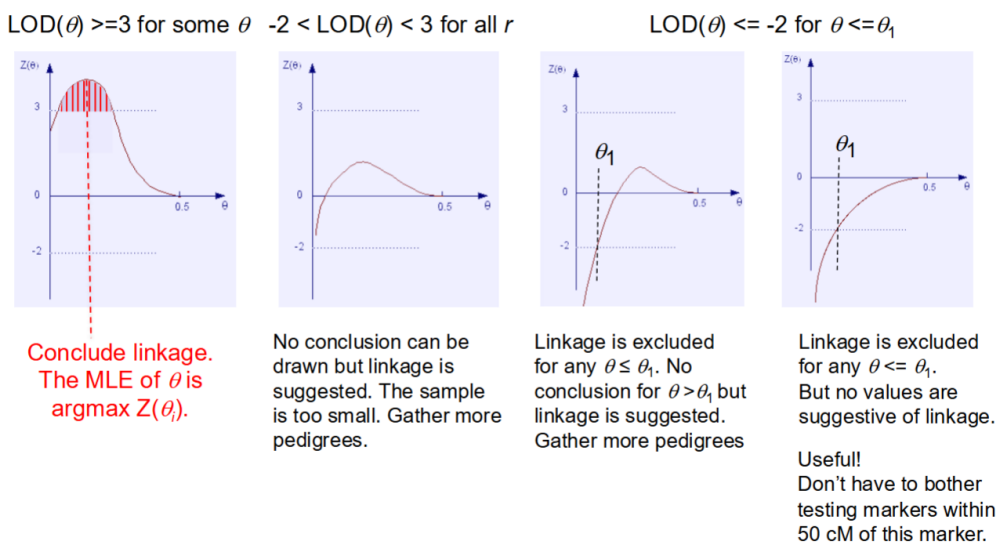
MLE of θ

The computational algorithms test not a single value of θ, but a range of values θ_i between 0 and 0.5 on each pedigree, adding the scores over all the pedigrees j to obtain:

$$Z(\theta_i) = \sum_j LOD(\theta_i)$$

If $Z(\theta_i) \geq 3$ for some θ_i, the maximum likelihood estimate of θ is $\arg\max Z(\theta_i)$

Summary of LOD Score Interpretation



Detection of Linkage in Human Pedigrees

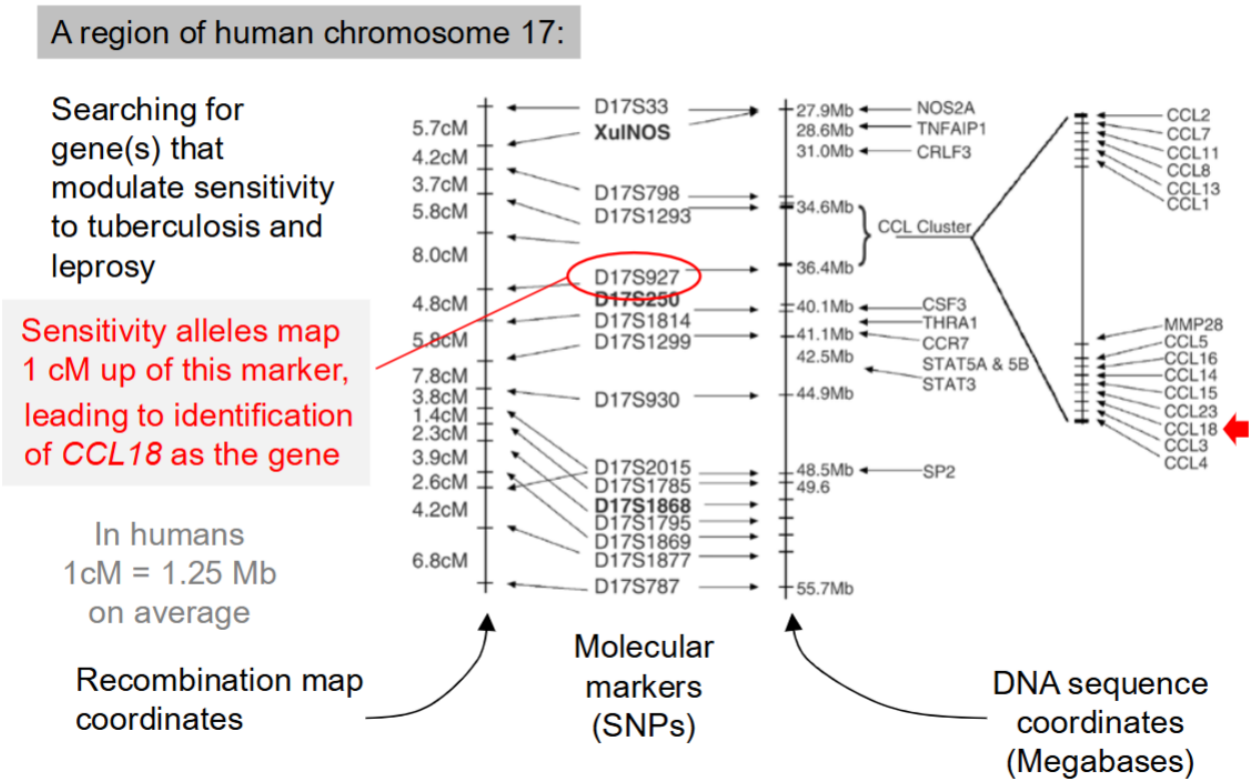
Major points:

- LOD Scores provide a simple metric to indicate whether a linked gene model explains a set of pedigree data better than an unlinked gene model.
- LOD scores reflect the size of the data set, so a higher score is achieved with a larger data set (if the linked gene model really explains the data). We can add LOD scores for new pedigrees as they become available.
- Our example was substantially simplified because
  - we knew the coupling relationship between markers
  - we had a perfect test-cross
  - complete dominance, penetrance, and expressivity
- Additional probabilities can be factored in, to account for deviations from this perfect situation. Computationally intensive.

Recombinational and Physical Chromosome Maps

- How do we use these types of maps?
- A physical map tells us where the trait determinants ("genes") lie and, from sequence analysis, what the gene products may do biochemically.
- A recombinational map tells us what a gene does biologically... and the chance that we have inherited a particular allele

Recombination Maps and Genome DNA Sequence are Combined to Identify Genes for Important Traits



Cross-over Frequency Varies Characteristically along Chromosomes

- Frequency is low at centromeres and telomeres
- As a result, if we look at recombination rate maps, there will be much lower recombination near centromeres and telmoeres and higher in the between
- This leads to "cold spots" and "hot spots"

Rate of Recombination Between Sexes

There are more recombination events in **females**.

How do we Find the Genes for Complex Traits

Known as "Quantitative Trait Loci" (QTLs). The problems we have to overcome:

1. Determined by combined effects of multiple genes
2. A given phenotypic value may correspond to many *different combinations of alleles* at multiple genes
3. Environmental effects can also blur the relationship between phenotypic value and genotype

Not as straightforward as recombination-mapping of single-gene Mendelian traits

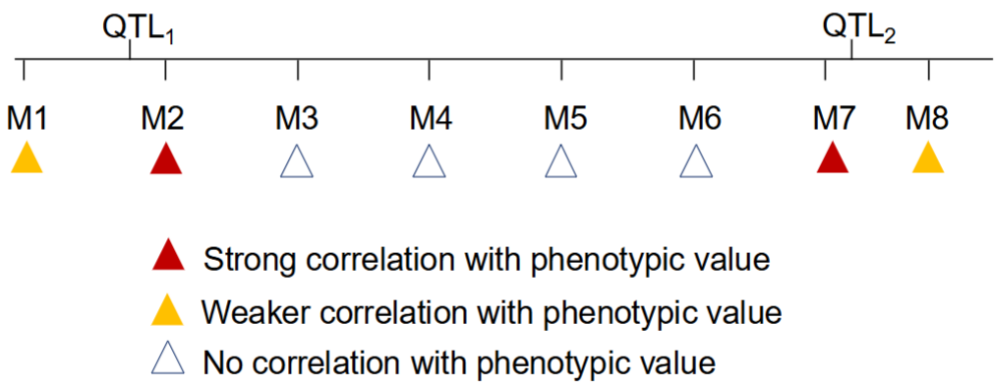
General Approach

Identify statistically significant correlations between

- phenotypic values for a particular trait
- genotypes at specific molecular marker loci

Large numbers of pre-selected markers throughout the genome are tested at the same time. The variants at the marker loci do not necessarily influence the trait (they usually don't) - they only serve to identify the approximate locations of the variants that do.

Example: M1 - M8 are tested markers



The observed correlations between phenotypic values and marker genotypes reveal the locations of QTL_1 and QTL_2.

How does Association Arise?

- The marker genotype itself influences the trait (low prior probability)
- The marker is closely linked to a sequence variant that influences the trait:
  - For **progeny of a specific mating:**
    - $1 - \theta$ = probability that the parental allele combination is passed on to an offspring (transmission bias)
  - For a **random-mating population**
    - **Linkage disequilibrium:** over short distances ($\theta << 0.5$), the ancestral allele combinations remain unchanged for many generations

Two Broad Types of Methods:

Both ultimately rely on linkage and recombination, but not on measurements of recombination frequency

1. Associations in populations with *known pedigrees*
   - Progeny of controlled crosses: experimental and agricultural organisms
   - Available human pedigrees: if large allele effect - transmission bias
2. Associations in *random-mating populations*
   - Known as GWAS: genome-wide association studies
   - Complex human traits with large or small allele effects

1. QTL Mapping Using Controlled Crosses

Example: Fruit weight in commercial tomato strains.

- First, identify many molecular markers that differ between the inbred parent strains (e.g., a SNP every 5 cM.)
- Suppose the Beefmaster tomato (mean 230g) has B/B alleles at all genes
- The Sungold tomato (mean 10g) has S/S alleles at all genes.
- Using the $n = \frac{D^2}{8V_g}$ formula, we predict there are 10 genes affecting weight. As a result, the average contribution per Beefmaster allele provides $(230 - 10)/10 = 22$g to weight.

Now, suppose we cross the Beefmaster and Sungold. The result is a heterozygous tomato at all genes. (B/S at all genes). Following this, we cross the F1 with one of the parents (say, the Beefmaster tomato in this case).

- The result is varying weights of tomatoes, with weights ranging from the same as the B/S F1 heterozygote, to the wild-type Beefmaster.
- Every SNP marker either has B/B or B/S, but none will have S/S.

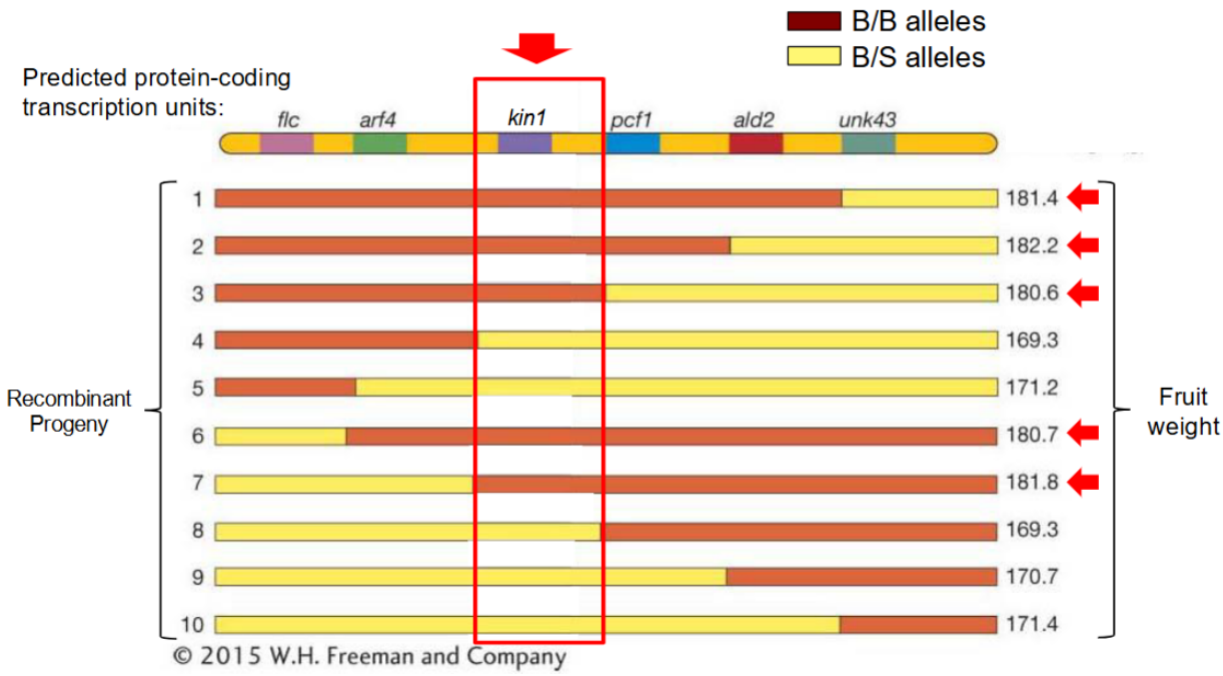
The result is the following:

		Markers						
Plant	**Fruit wt. (g)**	**M1**	***M2***	***M3***	***M4***	***M5***		
Beefmaster	230	B/B	B/B	B/B	B/B	B/B		Parent Lines
Sungold	10	S/S	S/S	S/S	S/S	S/S		
BC_1-001	183	B/B	B/B	B/B	B/S	B/S	R	Backcross Progeny (Parentals and Recombinants) But we don't care about their frequencies
BC_1-002	176	B/S	B/S	B/B	B/B	B/B	R	
BC_1-003	170	B/B	B/S	B/S	B/S	B/S	R	
BC_1-004	185	B/B	B/B	B/B	B/S	B/S	R	
BC_1-005	182	B/B	B/B	B/B	B/B	B/B	P	
BC_1-006	170	B/S	B/S	B/S	B/S	B/B	R	
BC_1-007	170	B/B	B/S	B/S	B/S	B/S	R	
BC_1-008	174	B/S	B/S	B/S	B/S	B/S	P	
BC_1-009	171	B/S	B/S	B/S	B/B	B/B	R	
BC_1-010	180	B/S	B/S	B/B	B/B	B/B	R	
BC_1-011	185	B/S	B/B	B/B	B/S	B/S	R	
BC_1-012	169	B/S	B/S	B/S	B/S	B/S	P	
BC_1-013	165	B/B	B/B	B/S	B/S	B/S	R	
BC_1-014	181	B/S	B/S	B/B	B/B	B/S	R	
BC_1-015	169	B/S	B/S	B/S	B/B	B/B	R	
BC_1-016	182	B/B	B/B	B/B	B/S	B/S	R	
BC_1-017	179	B/S	B/S	B/B	B/B	B/B	R	
BC_1-018	182	B/S	B/B	B/B	B/B	B/B	R	
BC_1-019	168	B/S	B/S	B/S	B/B	B/B	R	
BC_1-020	173	B/B	B/B	B/B	B/B	B/B	P	

We care about the genotype at each marker in each offspring, and the mean phenotypic value when the B or S allele is inherited.

- We can now use a statistical test to determine how much each individual gene contributes to the weight
- For example, using the above chart, if we compare the means of B/B and B/S at M3, we can determine it contributes around 11g.

We can define the QTL location more narrowly by analyzing more markers between M2 and M3, and then comparing with the location of the predicted genes in this region:



The QTL regions are further evaluated to identify the causal alleles and the relevant genes.

- The QTL region is mapped to between two markers
- More known markers between M_i and M_j are genotyped, OR, the DNA between those markers is sequenced from individuals with different phenotypes
- The causal variant and gene are identified based on:
  - Sequence analysis
  - Expression pattern
  - Evolutionary comparisons
  - Experimental tests

Finding QTLs in Humans

We cannot use the same approach because:

- We cannot perform controlled crosses
- No "true-breeding" humans
- Small numbers of progeny

Solution:

- Analyze genotype-phenotype associations in *random-mating* populations
- Approach known as GWAS (genome-wide association studies)

Concepts for GWAS

1. Molecular Markoers
2. Haplotypes
3. Linkage Disequilibrium
4. Haplotype Blocks

The latter three lead to a dramatic reduction of hypothesis search space. Every variant in the genome is a potential QTL hypothesis.

Choice of Molecular Markers

SNPs for humans: identified by the Thousand Genome and HapMap Projects

- Advantages:
  - Their exact physical locations in the genome are known.
  - Abundant and distributed uniformly throughout genome, so there is a high probability that **some** will be linked to the relevant genes.
  - Large numbers can be assessed at once by microarray hybridization
  - Genotypes (homozygotes, heterozygotes, hemizygotes) can be distinguished readily

Marker Methods

Marker **discovery** is by genome sequencing of many individuals

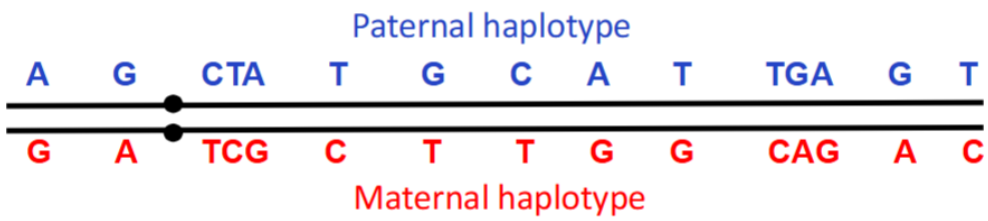
International HapMap Project

- discovers the variations by sequencing
- catalogues the variations and maintains database
  - location in the genome sequence
  - nature of change
  - frequency in population
  - haplotype combinations
- identifies a set of markers suitable for GWAS and mapping
  - common variations (minor allele frequency > 0.05)
  - reliability of genotyping
  - maximize coverage of genome
  - (1×10^6 SNPs for resolution $10000 - 100000$ bp)
- Marker **scoring** in cases vs controls is by microarray hybridization

Haplotypes

Haplotypes are combinations of syntenic alleles in an individual that were inherited together from one parent.

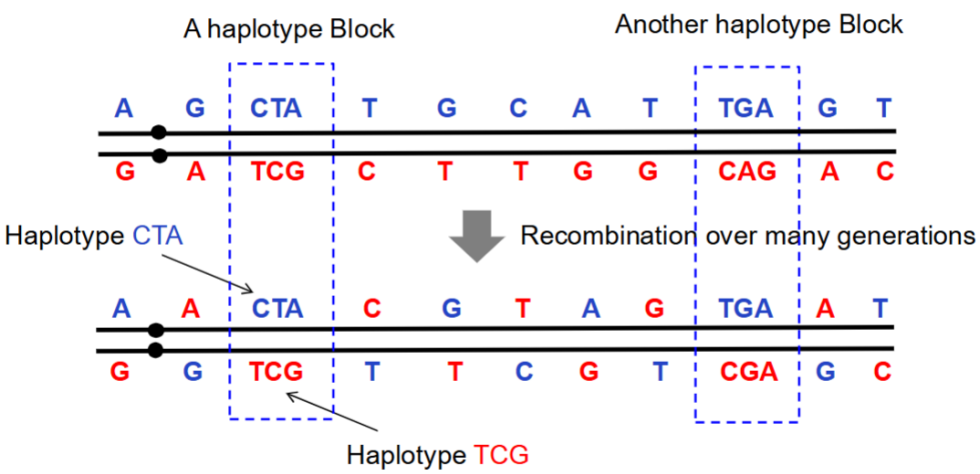
- e.g., for a particular pair of homologous chromosomes in a particular individual
- In other words, a haplotype is the genotype of an entire chromosome (or portion of a chromosome)
- These haplotypes are broken up/scrambled during crossing over



Linkage Disequilibrium

Some local haplotypes occur more frequently than expected at random

- Over time, combinations of alleles at distant loci on the same chromosome become randomized in the population, as a consequence of crossing over
- But alleles of loci that are close together tend to remain in the same combinations over time (stay in linkage disequilibrium)



Haplotype blocks are groups of alleles that are always inherited together due to linkage disequilibrium

Quantification of Linkage Disequilibrium

A useful metric is r^2, equivalent to the Pearson correlation coefficient. For two biallelic loci $A(a)$ and $B(b)$:

$$r^2(p_a, p_b, p_{ab}) = \frac{D^2}{p_a p_b p_A p_B} = \frac{(p_{ab} - p_a p_b)^2}{p_a p_b p_A p_B}$$

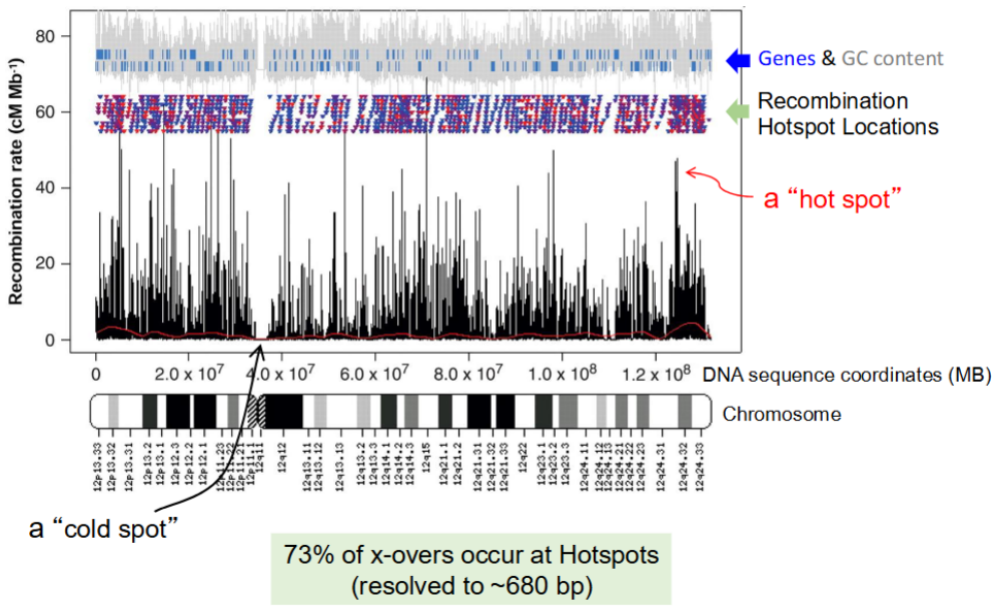
- p_x is the frequency of allele or haplotype x
- D is (observed − expected) haplotype frequency

The range for r^2 is from 0 to 1.

- $r^2 = 0$: no disequilibrium (random allele combinations)
- $r^2 > 0$: increasing disequilibrium (biased allele combinations)
- $r^2 = 1$: full disequilibrium (if minor allele frequencies are equal)

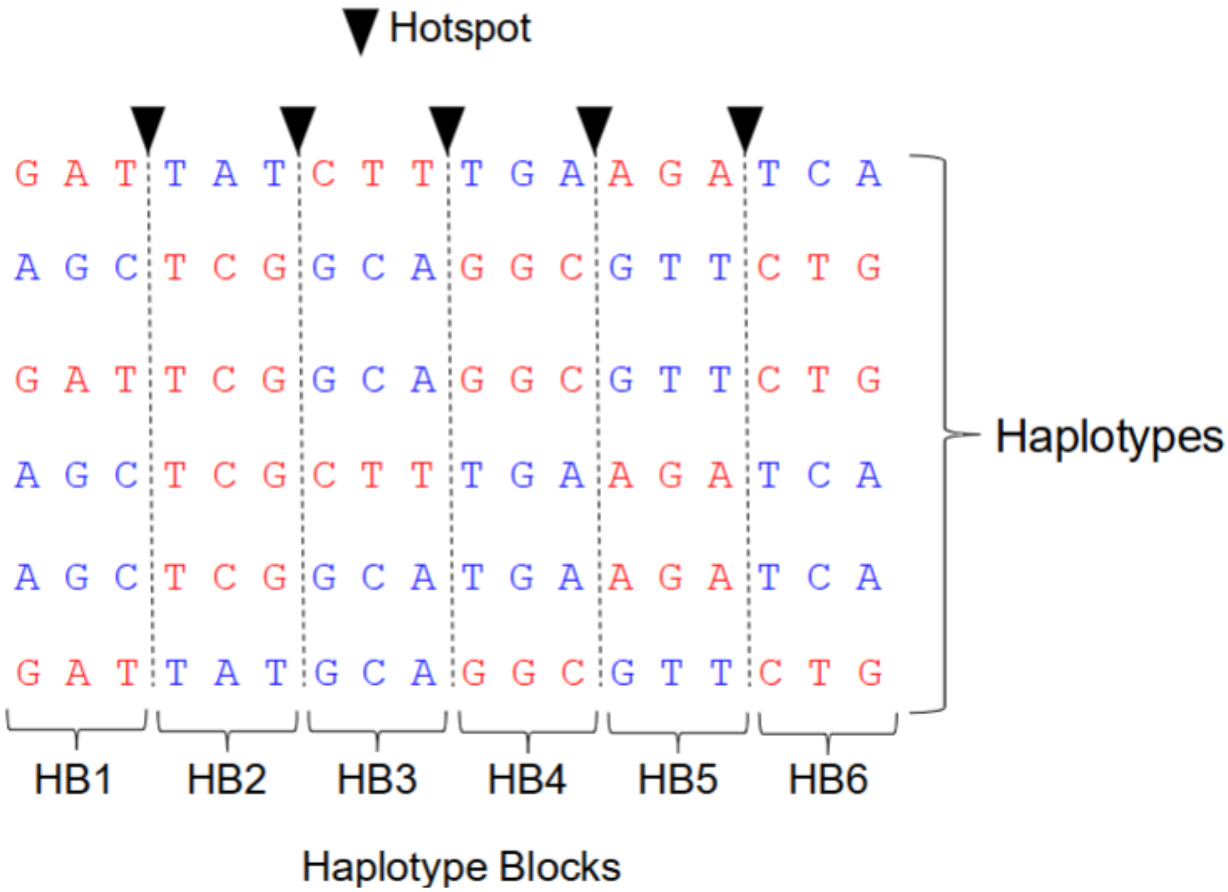
We'll see later that r^2 can be used to adjust estimates of statistical power and required sample sizes in genetic association studies.

Linkage Disequilibrium is Enhanced by Local Variations in Crossover Frequency



Halldorsson BV, et. al. (2019) *Science* 363: eaau1043

Recombination hotspots enhance linkage disequilibrium and give rise to haplotype blocks.



Haplotypes and Individual Genotypes

If two haplotypes exist in a population for loci $E(e)$ $F(f)$ $G(g)$ $H(h)$

- haplotype 1: $EFgH$
- haplotype 2: $efGh$

an individual can have one of these three genotypes

$$\frac{EFgH}{EFgH} \quad \text{or} \quad \frac{EFgH}{efGh} \quad \text{or} \quad \frac{efGh}{efGh}$$

If we only score the genotype for one marker, then we can infer (or "impute") the genotype for the other three:

- If genotype F/f is detected in an individual, it is possible to infer genotypes E/e, g/G, and H/h

Similarly, if $H(h)$ or an *unknown* variant site within this haplotype block affects a phenotype, the phenotypic value will correlate with the presence of F versus f allele.

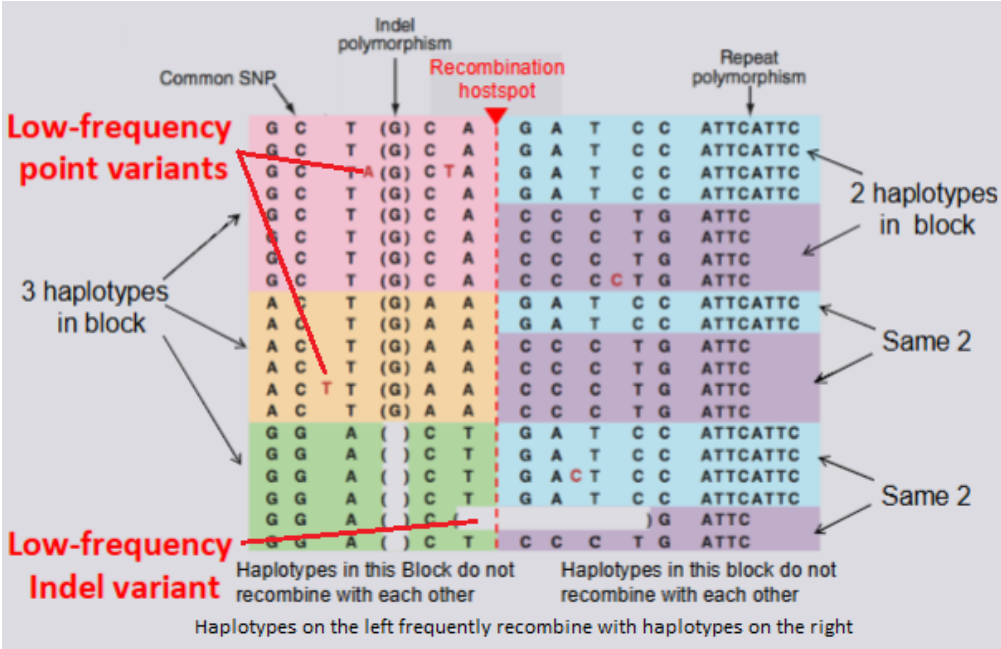
Hotspot Locations

Recombination hotspots tend to be located in **regulatory regions** (rather than within transcribed regions)

The following is an example of haplotypes across a small chromosomal region

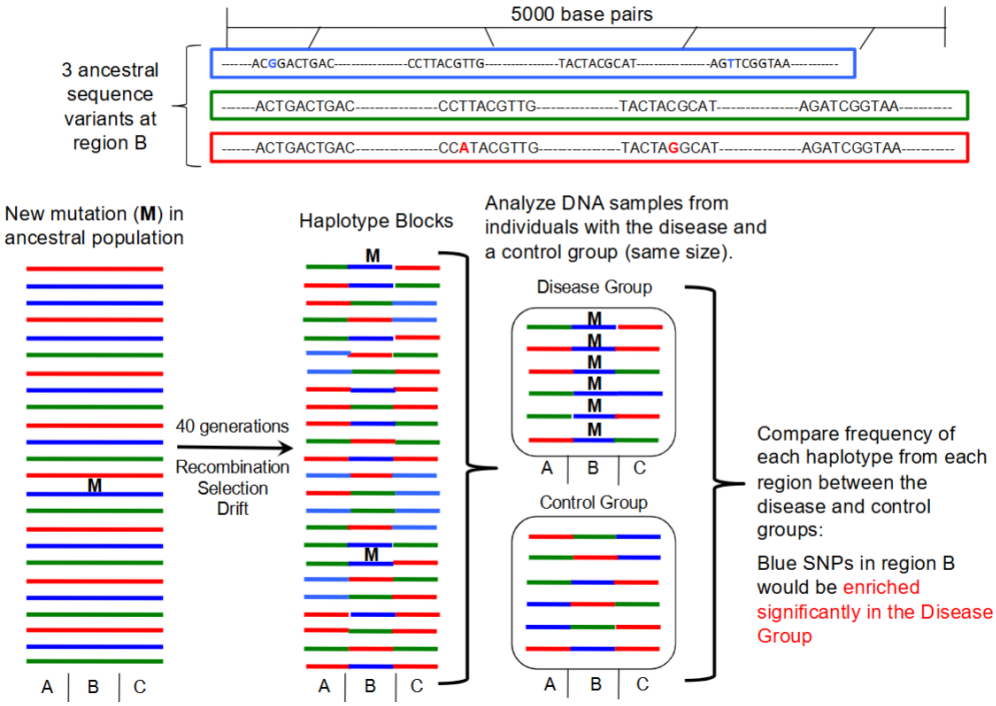
- only sequence variations are shown, not all the nucleotides

- blank spaces can represent **many** invariant nucleotide positions



Low-frequency variants that may be QTLs have arisen by mutation within specific haplotype blocks, so they remain associated with the haplotype of origin (they do not recombine into other haplotypes in the block)

GWAS Mapping of a Disease Risk Gene Using Haplotypes



Association between Continuous, Meristic, and Threshold Traits

For visible continuous or meristic traits, we would proceed with the same type of analysis as discussed earlier for Tomato fruit weight:

- Test for association between marker alleles and mean phenotypic value (but among individuals in a random-mating population)

For threshold traits, the relevant quantitative phenotype (the underlying liability) is hidden.

- We test instead for association with the emergent binary trait: affected/non-affected, as we'll describe now.

However, keep in mind that in some cases, we can also test for association with a continuous trait known to be a risk factor: e.g., serum cholesterol levels, insulin levels, ...

Effect Size Associated with a Marker

The Outcomes Odds Ratio

Ideally, we want the **Outcome Odds Ratio:** the odds of being affected given a particular genotype, divided by the odds given the absence of that genotype.

$$OOR = \frac{\text{Odds affected in exposed}}{Oddsaffectedinnon-exposed}$$

This requires pre-selecting samples of different genotypes, and then determining their phenotypes.

For late-onset traits, the Outcome Odds Ratio requires following the individuals over their lifetime.

The Exposure Odds Ratio

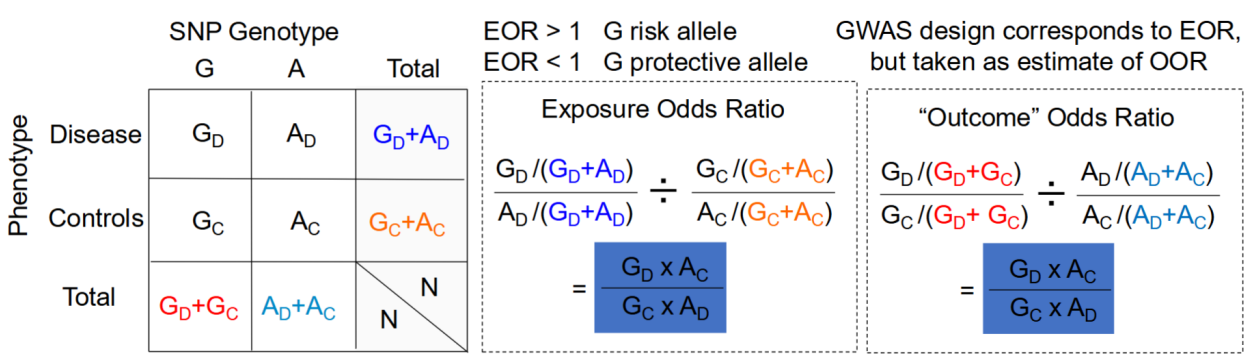
GWAS yield the **Exposure Odds Ratio**: the odds of carrying a specific genotype given that the disease state is affected, divided by the odds of carrying that same genotype given that the disease state is unaffected.

$$EOR = \frac{\text{Odds exposed in affected}}{\text{Odds exposed in non-affected}}$$

This is mathematically equilivant to OOR.

- In GWAS, equal-sized samples of affected and unaffected are collected, and their genotypes are then determined.

Odds Ratios



Confidence Intervals for the Odds Raio

- Upper 95% CI $= \exp\left(\ln(OR) + 1.96\sqrt{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}}\right)$
- Lower 95% CI $= \exp\left(\ln(OR) - 1.96\sqrt{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}}\right)$

If the confidence interval includes the value 1, the odds ratio is not considered significant

Significance of Association: χ^2 Test of Independence

For each SNP tested, SNP1, SNP2, ..., SNPn

- H_0 = same allele distribution in cases and controls
- H_a = different allele distribution in cases and controls

we find the values of the tables above, G_D, A_D, G_C, A_C, and their sums.

- This makes no assumption about the relative impact of homozygotes or heterozygotes: accommodates dominant/recessive, additive and multiplicative models with best generalized power and computational efficiency.

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

The expected frequencies are the following:

$$E(G_D) = (G_D + A_D)\frac{(G_D + G_C)}{N}$$
$$E(G_C) = (G_C + A_C)\frac{(G_D + G_C)}{N}$$
$$E(A_D) = (G_D + A_D)\frac{(A_D + A_C)}{N}$$
$$E(A_C) = (G_C + A_C)\frac{(A_D + A_C)}{N}$$

In this situation, we have $dof = 1$

Must Adjust for Testing of Multiple Hypotheses

Each SNP represents a separate hypothesis tested (SNP association with the trait)

- With $P \leq 0.05$ to reject H_0, we accept a 5% probability of incorrectly rejecting H_0 for the SNP tested (a false positive)

If we test 1×10^6 SNPs with threshold $P \leq 0.05$, we expect to accept

$$(1 \times 10^6)(0.05) = 5 \times 10^4 \text{ false positives!}$$

Therefore, we must raise the threshold for significance.

- A simple (but conservative) solution: **Bonferroni Correction**

$$\alpha^* = \alpha In$$

- α^* = adjusted threshold for a single test (SNP)
- α = overall false positive threshold (i.e., P of at least 1 false)
- n = number of tests (SNPs)

This is **not** the best solution, but it has been the standard for reporting results, leading to problems that we will discuss later.

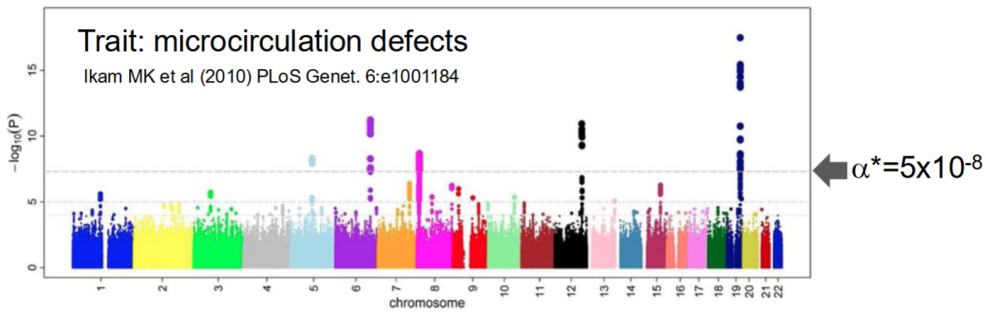
For the above example, if we want an overall false positive rate of 5% with 1×10^6 SNPs, we must set the threshold for individual SNPs to:

$$\alpha^* = (0.05)/(1 \times 10^6) = 5 \times 10^{-8}$$

This is the threshold most commonly used in GWAS for "genome-wide significance"

- It involves a **tradeoff**: we will reject many true positives. (Type II Error: false negatives)

Visualization of GWAS Results: "Manhattan" Plots



SNPS that are *correlated* (**not causated**) with the trait we are studying would have the "skyscrapers".

- Thousands of SNP associations have been identified for diverse human traits
- GWAS detects correlation, not causation. Relatively few cases have the relevant genes identified.

GWAS is only an **initial** low-resolution analysis.

- Uses common SNPs pre-selected to maximize coverage and statistical robustness with limited sample size
- Results point to relevant **neighborhoods**, but the regions can contain many genes
- The relevant alleles are frequently NOT the tested SNPs, but rather other genetic variations in linkage disequilibrium with them.

High-resolution follow-up is required.

Some Questions About GWAS Results

- How do we know whether we have identified all the QTLs?
  - How much of the measured heritability is actually explained by the identified loci?
- Can we predict phenotype (e.g., disease risk) from genotype?
- What do current results show about the genetic architecture of complex traits? (as opposed to long-standing hypotheses?)
- What are the broad implications for
  - evolution of complex traits
  - development of therapies for complex diseases

SNP-based Heritability: General

$$\text{PRS} = \text{Polygenic Risk Score} = \sum_{SNP_j} x_j \beta_j$$

- SNP_j is the set of tested SNPs
- x is the allele dosage at SNP j: 0, 1, or 2 copies of the trait-increasing allele
- β is the additive allele effect from the data (from odds ratio)

$$h^2_{PRS} = \frac{var\left(\sum_{SNP_j} x_j \beta_j\right)}{var(y)}$$

- y is the trait phenotypic value

If restricted to the set of SNPs that achieve genome-wide significance, then this heritability is sometimes called h^2_{GWAS}.

- If all the QTLs have been identified, h^2_{PRS} should be equal to the traditional estimates of narrow-sense heritability h^2.
- **Continuous traits**: comparison is straightforward because we can measure the SNP effect as well as V_P and V_A in the same real physical units.
- **Threshold traits**: comparison is more complicated because the quantitative trait is the underlying liability (for which we calculate $h^2_{liability}$ based on recurrent risk λ_s in the population). But what we get from the GWAS experiment is the SNP contribution to the **observed binary trait** (the probability of being affected, in a non-representative sample).

SNP-based Heritability: Threshold Traits

The standardized individual phenotype (e.g., probability of being affected) is:

$$y_j = m + \sum_{i \in C} z_{ij} \alpha_i + \epsilon_j$$

- z_{ij} is the standardized, normalized genotype at gene i: $(x_{ij} - 2p_i)/(2p_i q_i)^{\frac{1}{2}}$
  - x is the dosage at gene i, p and q are the allele frequencies in sample
  - $2p$ is the expected value (mean for a binomial distribution with 2 trials). (2 alleles per individual = 2 trials)
  - $2pq$ is the variance for a binomial distribution with 2 trials
- α is the effect size in standard deviation units
- ϵ is the environmental contribution

The SNP contribution to phenotypic variance is then: $\sigma^2_g = \sum_i \alpha^2_i$, and the narrow-sense heritability accounted for by the SNPs is:

$$h^2 = \frac{\sigma^2_g}{\sigma^2_g + \sigma^2_\epsilon} = \sigma^2_g = \sum_i \alpha^2_i$$

The conversion between the heritability of the observed binary trait (h^2_{PRS}) and the heritability of the continuous liability can be computed as follows:

$$h^2_{liability} = [h^2_{observed} \cdot \frac{K(1-K)}{\varphi(\phi^{-1}[K])^2}] \cdot \frac{K(1-K)}{P(1-P)}$$

- K is the frequency of the binary trait in the population
- P is the frequency of the binary trait in the observed sample
- the **denominator** of the first fraction is the squared height z of the standard normal probability density function at the liability threshold t (i.e., the squared probability density function evaluated at the K quantile of the inverse comuluative density function)

SNP-based Heritability

- In principle, if all the genes have been found, **an individual's PRS gives us the genetic contribution to their phenotypic value**.
- If we also understnad the environmental effects, **we can predict the phenotypic value**.
- If we have found all the genes, we expect

$$h^2_{PRS} = h^2_{measured}$$

- But we don't get that. This leads to the missing heritability problem.

Missing Heritability Problem

Thousands of SNP associations have been identified for diverse human traits, but the genome-wide significant SNPs only account for a **small proportion** of the measured heritability for any trait. Why?

- Over-conservative correction for multiple hypothesis testing
- The multi-gene architecture of complex traits (makes the over-correction problem worse)
- Possible overestimation from family studies
  - Shared / "private" environments
  - Hidden non-additive genetic effects

Over-conservative Multiple-Testing Correction

- Bonferroni is very conservative: many true positives will be rejected
- Makes incorrect assumption of marker independence:
  - Markers in the same haplotype block are correlated; they are associated with the same causal locus
  - Even markers in other nearby blocks can be correlated due to partial linkage

A better method: **Holm-Bonferroni Correction**

Holm-Bonferroni Correction

- More power than Bonferroni but equal protection against Type I error
- Ensures family-wise error rate (FWER) is controlled to $\leq \alpha$
- Does not assume independent hypotheses

To do this:

- Let $H_1, \ldots, H_m$ be a family of m null hypotheses and $P_1, \ldots, P_m$ the corresponding p-values

- Order the p-values (from lowest to highest) $P_{(1)}, \ldots, P_{(m)}$
- For a given significance level α, let k be the minimal index such that $P_{(k)} > \frac{\alpha}{m+1-k}$
- Reject the null hypotheses $H_{(1)} \ldots H_{(k-1)}$ and do not reject $H_{(k)} \ldots H_{(m)}$
- If $k = 1$, then do not reject **any** of the null hypotheses
- If no such k exists then reject all of the null hypotheses.

Genetic Architecture

The second factor behind hidden heritability. What do we actually expect to see in the case of a complex disease trait?

a) Rare single-gene Mendelian trait with complete penetrance

b) Rare allele contributing large effect to a complex trait

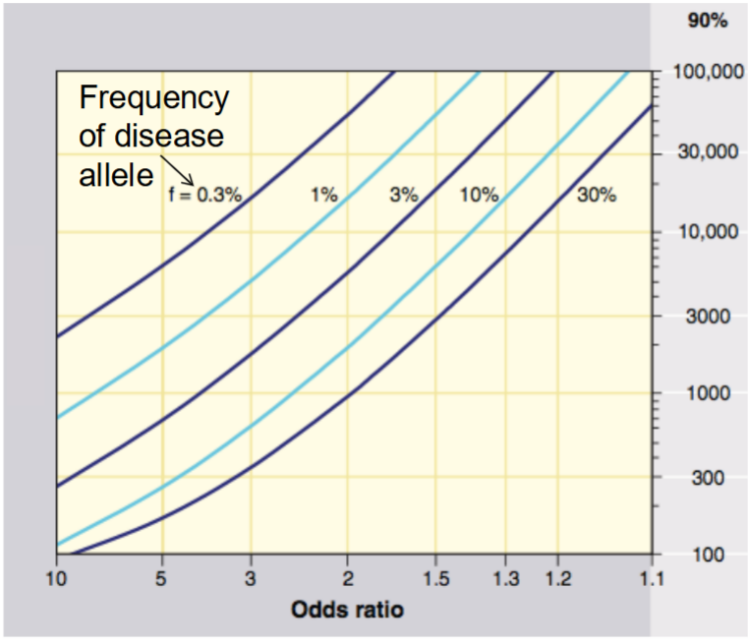
c) Common allele contributing small effect to a complex trait

Hint: It's the common allele.

Genetic Explanations of the "Missing Heritability" (not mutually exclusive)

- **Infinitesimal hypothesis**:
  - Proposes many common alleles at many genes with small effects each
  - Strongly supported by current evidence - updated to "Omnigene" Model
- **Rare allele hypothesis**
  - Proposes large effects by rare alleles (freq $< 1\%$)
  - Difficult to detect with standard-size studies
  - More being detected with increasing sample sizes, but little contribution to h^2
- **Epistasis hypothesis**
  - Proposes hidden epistasis hidden in heritability studies
  - Can be tested with the same data (compare Association for single SNPs versus combinations of SNPs)
  - But complex analysis and difficult to detect in standard-size studies
  - Problem of multiple hypothesis testing is magnified
  - not supported by current evidence (although examples exist)

Sample sizes required for GWAS



- x-axis: Odds ratio = magnitude of gene effect
- y-axis: Sample size needed for 90% probability to achieve a significant result at $P < 10^{-8}$ (infinitesimal model)

What is the actual genetic architecture of complex traits?

Ronald Fisher's Infinitesimal Model (1918): large number of genes, with common alleles each contributing a small additive effect.

- Does this model hold up in the face of modern GWAS evidence, especially for common threshold disease traits?
- How important are rare alleles with large effect?
- How important are non-additive effects?

Genetic Explanations of the "Missing Heritability" (not mutually exclusive)

- Infinitesimal hypothesis
  - proposes many common alleles at many genes with small effects each
  - **strongly supported by current evidence - updated to "Omnigene Model"**
- Rare allele hypothesis
  - proposes large effects by rare alleles (freq <1%)
  - difficult to detect with standard-size studies
  - more being detected with increasing sample sizes, **but little contribution to h^2**
- Epistasis hypothesis
  - proposes extensive epistasis hidden in heritability studies

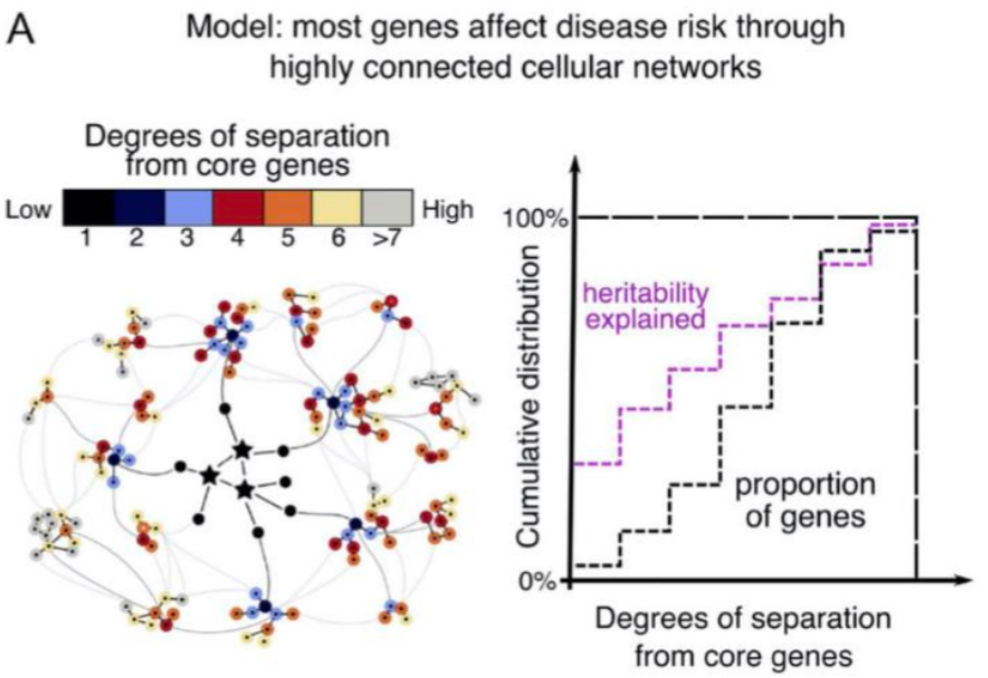
- can be tested with the same data (compare association for single SNPs versus combinations of SNPs)
- but complex analysis and difficult to detect in standard-size studies
- problem of multiple hypothesis testing is magnified
- **no supported by current evidence (although examples exist)**

Common variants (MAF >5%) have small additive effect sizes, but they are responsible for **most of the variance in the population**. In contrast, very rare variants can have large effects on disease, but they **contribute little to mean phenotype and variance** in the population.

Can we explain this in terms of the evolutionary processes of natural selection and adaptation?

1. Large deleterious effects will be strongly selected against (even if they can have beneficial effects in some contexts)
2. Spreading out effects among many common variants minimizes strong negative selection, maintaining a reservoir of diversity that allows strong positive responses when needed (recall the gene number effect)

An Expanded view of Complex Trait: From Polygenic to Omnigenic



*Boyle et al (2017).